

Female Infertility NGS Panel

Patient name	: Mrs. XX	PIN	: XXXXX
Gender/Age	: Female /32 Years	Sample number	: XXXX
Referring Clinician	: Dr. XXXX	Sample collection date	: 29-09-2025
Hospital/ Clinic	: XXXXXX	Sample received date	: 30-09-2025
Specimen	: Peripheral blood	Report date	: 26.10.2025

INDICATION FOR TESTING

Proband, Mrs. XX is non-consanguineously married. Her Antral Follicle Count (AFC) is 6-8. Laboratory investigations revealed FSH – 5.72, estradiol – 54.59, progesterone – 1.53, LH – 3.68 and TORCH – HIV, HCV, HPV were all non-reactive. The couple have a history of ART cycle with self-gametes where 12 oocytes were retrieved and 2 embryos were formed. In the FET, two blastocysts were transferred and resulted in implantation failure. Proband, Mrs. XX has been evaluated for pathogenic variations and polymorphisms.

RESULTS

FEMALE POLYMORPHISM ANALYSIS

List of polymorphic variants (SNP) identified:

Gene Name	RsID	Variant	Genotype	Zygosity	Observed Phenotype	Recommendations
AMH	rs10407022	c.146G>T	TT	Homozygous Polymorphism	Associated with reduced bioactivity of AMH, Associated with unexplained infertility [4-7].	Clinical correlation is strongly recommended
FSHR	rs6166	c.2039G>A	GG	Homozygous Wild Type	Association with Controlled Ovarian Stimulation [7-9].	May require higher gonadotropin doses during ovarian stimulation. Clinical

						correlation is strongly recommended [7-9].
<i>MTRR</i>	rs1801394	c.66A>G	AG	Heterozygous Polymorphism	Recurrent Pregnancy Loss, Fetal Neural Tube Defects [10-11].	Clinical correlation is strongly recommended.
<i>MTHFR</i>	rs1801133	c.665C>T	CT	Heterozygous Polymorphism	Recurrent Pregnancy Loss, Fetal Neural Tube Defects risk [12-16].	Clinical correlation is strongly recommended.

FEMALE FERTILITY SINGLE NUCLEOTIDE VARIATION ANALYSIS

No pathogenic or likely pathogenic variant causative of the reported phenotype was detected

Additional Variant(s)

List of significant carrier variants identified:

Gene	Region	Variant*	Allele Status	Disease	Classification*	Inheritance pattern	Clinical & Literature Evidence
<i>HBB</i> (-)	Exon 1	c.92G>C (p.Arg31Thr)	Heterozygous	Thalassemia, beta (OMIM#613985)	Likely Pathogenic	Autosomal Recessive	Clinvar: 15234 Pubmed: 20406103
<i>CUL7</i> [#] (-)	Exon 8	c.1886dupG (p.Pro630Serfs Ter50)	Heterozygous	3-M syndrome 1 (OMIM#273750)	Likely Pathogenic	Autosomal Recessive	-
<i>ADGRG1</i> (+)	Intron 4	c.487+1G>A (5' splice site)	Heterozygous	Cortical dysplasia, complex, with other brain malformations 14A, (bilateral frontoparietal) (OMIM#606854)	Likely Pathogenic	Autosomal Recessive	-

CUL7[#] gene is not covered in female infertility panel

*Genetic test results are based on the recommendation of American college of Medical Genetics [1-3].

No other variant that warrants to be reported for the given clinical indication was identified.

Recommendations

- The *HBB* gene has pseudogene in human genome.
- Sequencing the variant(s) in the parents and the other affected and unaffected members of the family is recommended to confirm the significance.
- Genetic counseling is recommended.

Methodology: Single Nucleotide Variation - Single Nucleotide Polymorphism

SNV analysis: DNA extracted from the blood, was used to perform targeted gene capture using a custom capture kit (for SNV analysis – 828 genes). The targeted libraries were sequenced to a targeted depth of 80 to 100X using SurfSeq 5000 sequencing platform. This kit has deep exonic coverage of all the coding regions including the difficult to cover regions. The sequences obtained are aligned to human reference genome (GRCh38.p13) using Sentieon aligner and analyzed using Sentieon for removing duplicates, recalibration and re-alignment of indels. Sentieon DNAscope has been used to call the variants. Detected variants were annotated and filtered using the VarSeq software with the workflow implementing the ACMG guidelines for variant classification. The variants were annotated using 1000 genomes (V2), gnomAD (V4.1.0,3.1.2,2.1.1), ClinVar, OMIM, dbSNP, NCBI RefSeq Genes. *In-silico* predictions of the variant was carried out using VS-SIFT, VS-PolyPhen2, PhyloP, GERP++, GeneSplicer, MaxEntScan, NNSplice, PWM Splice Predictor. Only non-synonymous and splice site variants found in the coding regions were used for clinical interpretation. Silent variations that do not result in any change in amino acid in the coding region are not reported.

SNP analysis: Variant analysis (for SNP analysis – 30 genes) and interpretation is done using VarSeq Software. Extensive scientific literature, Information from variant analysis and disease specific databases, population specific research are used to interpret and recommend. All results are finally approved by medical geneticists.

Sequence data attributes

Total reads generated	12.95 Gb
Data ≥ Q30	96.63%

Genetic test results are reported based on the recommendations of American College of Medical Genetics [1], as described below:

Classification	Interpretation
Pathogenic	A disease-causing variation in a gene which can explain the patients' symptoms has been detected. This usually means that a suspected disorder for which testing had been requested has been confirmed
Likely Pathogenic	A variant which is very likely to contribute to the development of disease however, the scientific evidence is currently insufficient to prove this conclusively. Additional evidence is expected to confirm this assertion of pathogenicity.
Variant of Uncertain Significance	A variant has been detected, but it is difficult to classify it as either pathogenic (disease causing) or benign (non- disease causing) based on current available scientific evidence. Further testing of the patient or family members as recommended by your clinician may be needed. It is probable that their significance can be assessed only with time, subject to availability of scientific evidence.

Disclaimer

- The classification of variants of unknown significance can change over time. Anderson Diagnostics and Labs cannot be held responsible for it.
- Intronic variants, UTR, Promoter region variants and CNV are not assessed using this assay.
- Certain genes may not be covered completely, and few mutations could be missed. Variants not detected by this assay may impact the phenotype.
- The variations have not been validated by Sanger sequencing.
- The above findings and result interpretation was done based on the clinical indication provided at the time of reporting.
- It is also possible that a pathogenic variant is present in a gene that was not selected for analysis and/or interpretation in cases where insufficient phenotypic information is available.
- Genes with pseudogenes, paralog genes and genes with low complexity may have decreased sensitivity and specificity of variant detection and interpretation due to inability of the data and analysis tools to unambiguously determine the origin of the sequence data in such regions.
- Incidental or secondary findings that meet the ACMG guidelines can be given upon request [3].
- Result interpretation was done based on the literature evidence available at the time of reporting. The clinical significance of the polymorphic variants tested can change over time and Anderson Diagnostics & Labs cannot be held responsible for this.

- This is not a diagnostic test and so not to be considered as diagnosis of any disease. This test is meant only for understanding the polymorphism at a given position and its association with various clinical parameters.

References

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